

**Project Management Framework**

**1. Project Governance**

- Project Steering Committee
- Project Director
- Project Management Office (PMO)
- Decision-making matrix
- Escalation mechanism
- Stakeholder governance

**2. Project Planning**

- Project Execution Plan (PEP)
- Work Breakdown Structure (WBS)
- Master Schedule (Primavera/MS Project)
- Milestone planning
- Resource planning
- Budget planning

**3. Scope Management**

- Clearly define project boundaries.
- Scope freeze after approval.
- Change management process.
- Management of scope creep if required.

**4. Engineering Management**

- Design Basis Report
- Basic Engineering
- Detailed Engineering

- 3D Model Reviews
- HAZID/HAZOP studies
- Value Engineering
- Design approvals

#### **5. Procurement Management**

- Procurement strategy
- Bid package preparation
- Vendor prequalification
- Technical evaluation
- Commercial evaluation
- Contract award
- Expediting
- Inspection and testing

#### **6. Construction Management**

- Site mobilization
- Temporary facilities
- Construction sequencing
- Civil, mechanical, electrical, and instrumentation coordination
- Daily progress monitoring
- Productivity improvement

#### **7. Marine Construction Management**

- Marine logistics
- Dredging coordination
- Jetty construction
- Piling works
- Dolphin installation

- Marine equipment mobilization
- Tide and weather planning

#### **8. HSE Management**

- Zero Harm philosophy
- HSE Plan
- Permit to Work
- Behaviour-Based Safety
- Emergency Response Plan
- Environmental management
- Safety audits and inspections

#### **9. Quality Management**

- Quality Assurance Plan
- Inspection & Test Plans (ITP)
- Material inspection
- Welding quality
- NDT
- Factory Acceptance Tests (FAT)
- Site Acceptance Tests (SAT)

#### **10. Cost Management**

- CAPEX control
- Budget monitoring
- Cost forecasting
- Cash flow management
- Earned Value Management (EVM)
- Cost optimization

#### **11. Schedule Management**

- Baseline schedule
- Critical Path Method (CPM)
- Milestone tracking
- Recovery plans
- Delay analysis
- Progress reporting

## **12. Risk Management**

- Risk Register
- Technical risks
- Commercial risks
- Construction risks
- Environmental risks
- Regulatory risks
- Mitigation plans

## **13. Stakeholder Management**

- Consultants
- Contractors
- Government authorities
- Local communities
- Regulatory agencies
- Utility providers

## **14. Regulatory & Statutory Compliance**

## **15. Digital Project Controls**

- Primavera P6
- BIM/3D models

- Digital document management
- Drone monitoring
- GIS mapping
- Progress dashboards
- AI-assisted reporting (where appropriate)

#### **16. Testing & Commissioning**

- Pre-commissioning
- Mechanical completion
- Hydrotesting
- Flushing and cleaning
- Loop checking
- Functional testing
- Performance Guarantee Tests

#### **17. Handover & Close-Out**

- Punch list closure
- As-built drawings
- O&M manuals
- Training
- Spare parts handover
- Warranty management
- Project close-out report

#### **18. Post-Project Review**

- Lessons learned
- Performance evaluation
- Cost and schedule variance analysis
- Best practices

- Continuous improvement

**Key KPIs to be adhered.**

- Schedule Performance Index (SPI)
- Cost Performance Index (CPI)
- Engineering progress (%)
- Procurement progress (%)
- Construction progress (%)
- Commissioning progress (%)
- HSE performance (TRIR, LTIFR)
- Quality NCRs
- Change orders
- Risk closure rate
- Mechanical completion
- Ready for start-up (RFSU)
- Overall project completion (%)